DialogStudio: Towards Richest and Most Diverse Unified Dataset Collection for Conversational AI

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Abstract

Despite advancements in conversational AI, language models encounter challenges to handle diverse conversational tasks, and existing dialogue dataset collections often lack diversity and comprehensiveness. To tackle these issues, we introduce DialogStudio: the largest and most diverse collection of dialogue datasets, unified under a consistent format while preserving their original information. Our collection encompasses data from open-domain dialogues, task-oriented dialogues, natural language understanding, conversational recommendation, dialogue summarization, and knowledge-grounded dialogues, making it an incredibly rich and diverse resource for dialogue research and model training. To further enhance the utility of DialogStudio, we identify the licenses for each dataset, design external knowledge and domain-aware prompts for selected dialogues to facilitate instruction-aware fine-tuning. Furthermore, we develop conversational AI models using the dataset collection, and our experiments in both zero-shot and few-shot learning scenarios demonstrate the superiority of DialogStudio. To improve transparency and support dataset and task-based research, as well as language model pre-training, all datasets, licenses, codes, and models associated with DialogStudio are made publicly accessible[1].

1 Introduction

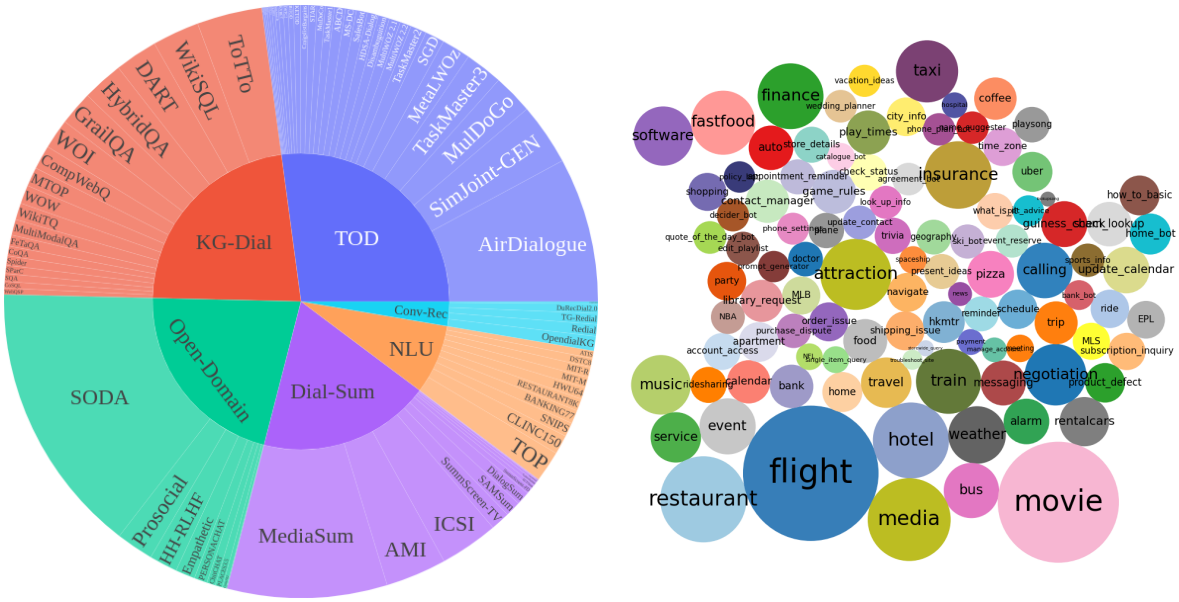
Recent years have seen remarkable progress in Conversational AI, primarily driven by the advent of approaches and language models (Shuster et al., 2022; Zhang et al., 2023; Longpre et al., 2023; Touvron et al., 2023). Despite the advancements, these models could fall short when handling various tasks in a conversation due to the lack of comprehensive and diverse training data. Current dialogue datasets (Lin et al., 2021; Asri et al., 2017) are typically limited in size and task-specific, which thus results in suboptimal ability in task-oriented model performance. Additionally, the lack of dataset standardization impedes model generalizability.

A few recent works (Gupta et al., 2022; Longpre et al., 2023; Ding et al., 2023) have introduced a large collection of datasets, which includes diverse tasks based on public datasets. For instance, FlanT5 (Longpre et al., 2023) presents the flan collections with a wide array of datasets and tasks. Despite this breadth, the coverage of dialogue datasets within the Flan collection remains notably sparse, featuring only about ten datasets. Although OPT (Iyer et al., 2022) have incorporated collections with several dialogue datasets, these collections remain inaccessible to the public. In contract, efforts like InstructDial (Gupta et al., 2022) and ParlAI (Miller et al., 2017) consist of more dialogue datasets, but they lack diversity and comprehensiveness. For instance, ParlAI mainly includes open-domain dialogue datasets, which are exclusively accessible through their platform. Other collections (Gupta et al., 2022; Kim et al., 2022a; Ding et al., 2023; Dubois et al., 2023) often distill single dataset from ChatGPT or process datasets into a sequence-to-sequence format to support language model training, featuring only input-output pairs such as dialogue context and system response. However, previous collections often overlook other crucial dialogue information, constraining their utility for research on individual datasets, tasks, and broader applications.

To overcome the aforementioned challenges, we introduce DialogStudio, the most comprehensive and diverse collection of publicly available dialogue datasets, unified under a consistent format. By aggregating dialogues from various sources, DialogStudio promotes holistic anal-

[*] Core contributors. Work completed during Kun's internship at Salesforce. Zhiwei is also a major contributor.

[1] https://github.com/salesforce/DialogStudio



(a) Dataset Distribution

(b) Domain Coverage of TOD

Figure 1: (a) is the distribution of all datasets in DialogStudio. The outer and inner circle list names of datasets and the associated categories, respectively. (b) illustrates covered domains of Task-Oriented Dialogues in DialogStudio.

ysis and the development of models adaptable to a variety of conversational scenarios. The collection spans an extensive range of domains, aspects, and tasks, and it is inclusive of several categories: Open-Domain Dialogues, Task-Oriented Dialogues, Natural Language Understanding, Conversational Recommendation, Dialogue Summarization, and Knowledge-Grounded Dialogues. Thus, it can provide support for research in both individual dialogue tasks and large-scale language pre-training.

DialogStudio stands out not only for its comprehensive coverage but also for its accessibility. It offers easy access with a unified format and documents. A straightforward `load_dataset()` command through HuggingFace allows users to seamlessly interact with the collection, and we have included documentation for each dataset to enhance usability. We anticipate that this collection will enable comprehensive and standardized training and evaluations of dialogue models, fostering fair comparisons and propelling further advancements in Conversational AI.

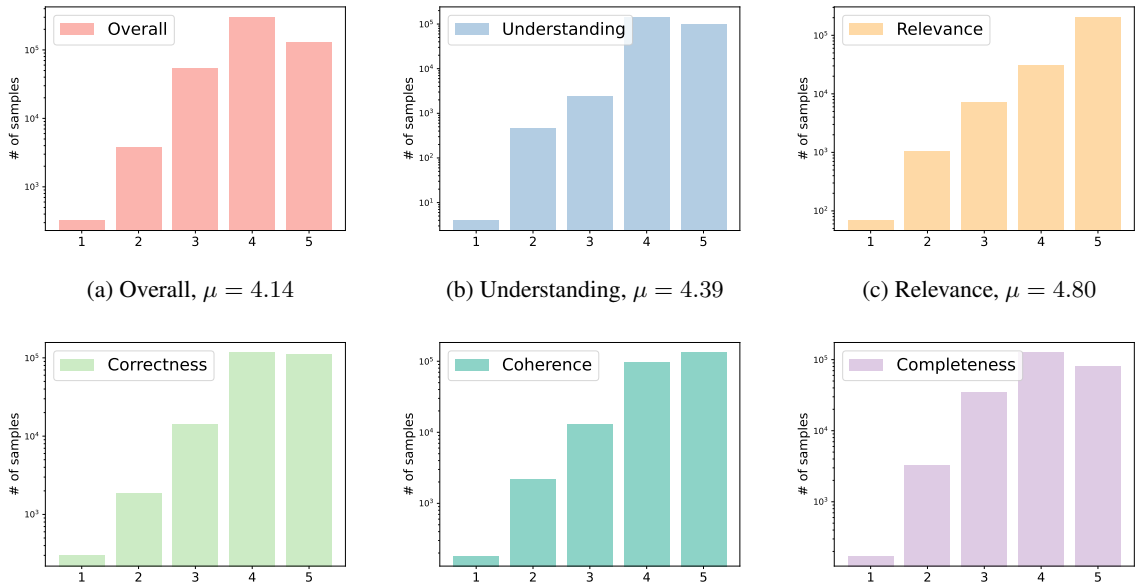
Furthermore, we identify dialogue domains, design external knowledge for available dialogues and create tailored prompts for selected datasets accordingly. Leveraging these datasets from DialogStudio, we have constructed instruction-aware models, with capacities ranging from 770M to 3B parameters. These models have the ability to handle various external knowledge and are adept at both response generation and general tasks, demonstrating the benefits of DialogStudio. The main contributions of this paper are as follows:

- We introduce DialogStudio, a meticulously curated collection of more than 80 dialogue datasets. These datasets are unified under a consistent format while retaining their original information. We integrate external knowledge, incorporate domain-aware prompts and identify dataset licenses, making DialogStudio an exceptionally rich and diverse resource for dialogue research and model training.
- We have made our datasets publicly available to enhance transparency and support research efforts. Additionally, we are committed to improving DialogStudio's usability and will persist in our efforts to refine it, ensuring an optimal user experience.
- We train conversational AI models based on DialogStudio, and these models have demonstrated superior performance over strong baselines in both zero-shot and few-shot learning scenarios.

2 Data analysis

2.1 Data Visualization

The dialogue datasets are compartmentalized into several categories: *Open-Domain Dialogues*,



(a) Overall, $\mu = 4.14$ (b) Understanding, $\mu = 4.39$ (c) Relevance, $\mu = 4.80$

(d) Correctness, $\mu = 4.38$ (e) Coherence, $\mu = 4.47$ (f) Completeness, $\mu = 4.16$

Figure 2: The score distribution for the dialogue quality.

Task-Oriented Dialogues (TOD), *Natural Language Understanding Dialogues (NLU)*, *Conversational Recommendation (Conv-Rec)*, *Dialogue Summarization (Dial-Sum)*, and *Knowledge-Grounded Dialogues (KG-Dial)*. Figure 1a presents an overview of DialogStudio's dataset categories. Note that the category boundaries are fuzzy as some datasets span multiple categories. For instance, SalesBot (Chiu et al., 2022) contains both casual and task-oriented conversations. Analogously, MultiWOZ (Budzianowski et al., 2018; Zang et al., 2020), a task-oriented dialogue corpus, incorporates knowledge bases and dialogue acts to enhance knowledge-grounded generation. Additionally, DialogStudio demonstrates its diversity by covering a wide range of domains, part of which is shown in Figure 1b.

2.2 Data Quality Investigation

Due to the existence of noise in dialogue, we develop a simple yet effective way to verify the quality of the datasets. Specifically, we employ ChatGPT (GPT-3.5-turbo) to evaluate the quality of system responses based on severall perspectives (Mehri et al., 2022; Kim et al., 2022a), *i.e.*, Understanding, Relevance, Correctness, Coherence, Completeness and Overall quality. Understanding assesses whether the model's responses accurately reflect the meaning and intent of the user's inputs. Relevance demonstrates whether the generated response should be directly related and appropriate to the preceding user input and the context of the conversation. Coherence measures the logical consistency of the model's responses within the context of the conversation. Completeness refers to whether the system's responses fully address the user's queries or tasks. Overall quality comprehensively rates the quality of dialogue. All scores are in the range of 1-5, and higher scores should only be given to truly exceptional examples. We delicately design the prompt and ask the ChatGPT model to *strictly* rate the score.

Since there are a lot of datasets in DialogStudio, we randomly select 33 multi-turn dialogue datasets and evaluate all the training dialogues of each dataset. To harmonize ChatGPT and human ratings, we take a random sample of 50 training dialogues from each dataset. These were then rated by three expert researchers using the five specified criteria. Post-alignment of ChatGPT and human evaluations, we view dialogues with a score above 3 as being of high quality. Figure 2 illustrates distributions of those scores. We also reveal the average score as the μ in each sub-caption. In general, the dialogues show high qualities regarding to the individual criteria and the overall quality.

3 Datasets Unification and Access

We collect and process a wide range of datasets, involving different domains, types, and tasks.

Since these datasets originally contain various information and format, we propose a unification strategy to process all the datasets such that they can be loaded in the same data loader.

3.1 Unification

Before unifying the format of those datasets, we fixed several issues as follows: 1) we remove those dialogues labeled as multi-turn dialogues, but actually with only one turn and miss either user utterance or system utterance. 2) We manually check the individual dialogues. If one dialogue contains one or more empty user or system utterances, we fill utterances based on corresponding dialogue contexts, dialogue acts, and dialogue information. In total, less than 0.5% of dialogues had these issues. To support research interest on individual datasets, we have flagged and rectified these problematic dialogues.

Additionally, we recognize the success of instruction tuning for dialogue models and thus we manually pre-define five different prompt templates for multi-turn dialogue datasets, such as *This is a bot helping users to* {*Task_Domain*}. *Given the dialogue context and external database, please generate a relevant system response for the user.* The {*Task_Domain*} is associated with the dialogue domain and we manually create a corresponding description. For example, if a dialogue is of domain *travel*, we set {*Task_Domain*} as *book a trip*. A concrete example of the prompt is demonstrated in Figure 3. Moreover, many datasets lack a direct mapping between dialogues and their domain information. To address this, we determine the domain of each dialogue using its intent, schema, APIs, and associated databases.

Next, we construct a uniform JSON dictionary format to store all relevant information of each dialogue as illustrated in Figure 3. Compared with existing works, DialogStudio covers more dialogue information and is easier to retrieve the information for arbitrary dialogue-related tasks. Concretely, we include all dialogue-related information, such as the dialogue ID, data split label, domain, task, and content. Additionally, we identify the external knowledge, dialogue state tracking (DST) knowledge, and intent knowledge in the dialogue, which are the most beneficial knowledge for a dialogue.

Regarding external knowledge, we construct it based on information such as databases and dialogue acts. Since each dialogue dataset focuses on specific tasks or domains and has a different database and annotation schema, we unify such information into *external knowledge*. For example, if the user is looking for a hotel and asking for its address, the system response should be based on both the search results from the database and the dialogue context. To simulate the realistic situation and avoid directly providing the model with the ground truth resulting hotel, we also randomly sample four other candidate results and mix them with the ground truth result. All information is flattened and converted into a string as external knowledge.

To complete tasks and generate coherent responses, a dialogue system needs to track users' requirements for the task. Those requirements are usually represented as dialogue states. For example, regarding the hotel booking task, a dialogue system needs to extract information such as price range and locations to enable searching hotels in the database. The type of dialogue states varies across different tasks and datasets. As such, it is hard for dialogue systems to predict the values of those dialogue states if unknowing the specific dialogue states the task covers. Therefore, we propose to insert the schema, consisting of predefined dialogue state types and values for each task, into the input sequence. For datasets like SGD (Rastogi et al., 2020), which already provides annotation schema, we directly convert the dictionary-structured schema into a string. For the rest datasets that have no such schema file, we iterate over all dialogues and collect potential state annotations to construct a schema. We provide domains, slot types, and slot values in the schema string. For those categorized dialogue slots like "hotel star-ratings", which have a fixed number of candidate values, we provide all possible values. For others that have unlimited possible values, e.g. "stay night", we randomly sample ten values, such that a model can learn what slot values are relevant to these slot types. We put the turn-level ground-truth DST information in "dst", and the general DST information under "dst knowledge", as presented in Figure 3.

Analogously, intent prediction also requires models to know all possible intent types for each task. Therefore, we extract the schema directly from the schema file if it exists. As to datasets without schema, we also iterate over all dialogue

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"dialogue_id": "train_1",
"num_utterances": 14,
"utterances": [
  {
    "speaker": "USR",
    "text": "I'd like to book a trip to Atlantis from Caprica on
            Saturday, August 13, 2016 for 8 adults.",
    "ap_label": "",
    "da_label": "inform"
  },
  {
    "speaker": "USR",
    "text": "I have a tight budget of 1700.",
    "ap_label": "",
    "da_label": "inform"
  },
  {
    "speaker": "SYS",
    "text": "Hi...I checked a few options for you, and we do
            not currently have any trips that meet this criteria.",
    "ap_label": "",
    "da_label": "sorry",
    "slots": {
      "dst_city": "Atlantis",
      "or_city": "Caprica",
      "str_date": "Saturday, August 13, 2016",
      "n_adults": "8",
      "budget": "1700"
    }
  }
 ],
"scenario": {
  "db_id": "U22HTHYNP",
  "db_type": "booking",
  "task": "book"
 }
```


(a) Original Data


```
"FRAMES--train--1": {
    "original dialog id": "train_1",
    "dialog index": 1,
    "original dialog info": {
      "scenario": {
        "db_id": "U22HTHYNP",
        "db_type": "booking",
        "task": "book"}}
   "log": [
      {
        "turn id": 1,
        "user utterance": "I'd like to book a trip to Atlantis from Caprica on Saturday,
                           August 13, 2016 for 8 adults. I have a tight budget of 1700.",
        "system response": "Hi...I checked a few options for you, and we do not currently
                           have any trips that meet this criteria.",
        "dialog history": "",
        "original user side information": {
          "da_label": "inform"
        },
        "original system side information": {
          "da_label": "sorry",
          "slots": {
            "dst_city": "Atlantis",
            "or_city": "Caprica",
            "str_date": "Saturday, August 13, 2016",
            "n_adults": "8",
            "budget": "1700"
          }
        },
        "intent": "inform",
        "dst": "book dst_city Atlantis, book or_city Caprica, book str_date Saturday, August
                13, 2016, book n_adults 8, book budget 1700"
      }
   ]
   "external knowledge": "( travel : (( trip : ( returning : ( duration : ( hours : 0 | min : 51...",
   "dst knowledge": " ( book : ( dst_city : ( Indianapolis | St. Loius | Le Paz | …) | or_city : (
                    PUebla | sf | toluca | San Francisco…",
   "intent knowledge": "( book : ( null | negate | request | goodbye | affirm))…",
   "prompt": [
      "This is a bot helping users to book a trip. Given the dialog context and external
       database, please generate a relevant system response for the user."
   ]
}
```


(b) DialogStudio Data

Figure 3: A dialogue format example. Left: original example, right: converted example. Here we only show the first turn and partial information.

in the dataset to collect all potential intents. Then, we put the turn-level ground-truth intent information into ”intent”, and the general intents under ”intent knowledge”, as presented in Figure 3. Note that not all datasets provide detailed annotation for dialogue states, intents, or even databases. For dialogue state tracking and intent classification tasks, we only process dialogues with corresponding annotations. Since all data is used for response generation, we leave the external knowledge value for the database blank if there is no related database in the original dataset.

3.2 Access and Maintenance

As aforementioned in the format, our DialogStudio data is easy to access via the JSON files. To make DialogStudio more maintainable and accessible, we will publish datasets on both GitHub and HuggingFace. GitHub mainly stores selected dialogue examples and relevant documents. We sample five original dialogues and five converted dialogues for each dataset to facilitate users in comprehending our format and examining the contents of each dataset. The complete DialogStudio dataset is maintained in our HugginFace repository, where all the datasets can be directly downloaded or loaded with the HuggingFace `load_dataset(dialogstudio, dataset_name)` API. Given the substantial volume of datasets, optimizing user experience poses a challenge and limitation. We will continuously maintain and update both GitHub and HuggingFace. DialogStudio is built upon public research datasets without individual or private information. We believe it is important to clearly present the license associated with each of these datasets. Consequently, we have included the original licenses for all datasets. All these datasets are supportive of academic research, and some of them also endorse commercial usage. The code that we employ falls under the widely accepted Apache 2.0 license. While we strictly require adherence to the respective dataset licenses for all intended usages on DialogStudio, there remains

a possibility that some works might not fully comply with the licenses.

Regarding the other concerns such as ethical concern, we admit that DialogStudio is collected and maintained by the authors of this work and we did not hire external annotators. Since it contains unified datasets across several categories, it supports various research purposes from individual tasks and datasets to language model pre-training.

4 Experiments

In this section, we present the pre-training details, methodologies, and metrics used to assess the performance of our DialogStudio model. The evaluation process aims to measure the model's ability to both solve task-oriented dialogues and understand general prompt-based instruction.

4.1 Model Pre-training

In this section, we introduce more details about how we conduct our pre-training. In regards of training models, we mix several datasets from DialogStudio.

For task-oriented and conversational recommendation datasets, we selected dialogues from a range of sources including KVRET (Eric et al., 2017), AirDialogue (Wei et al., 2018), DSTC2-Clean (Mrkšić et al., 2017), CaSiNo (Chawla et al., 2021), FRAMES (El Asri et al.), WOZ2.0 (Mrkšić et al., 2017), CraigslistBargains (He et al., 2018), Taskmaster1-2 (Byrne et al., 2019), ABCD (Chen et al., 2021a), MulDoGO (Peskov et al., 2019), BiTOD (Lin et al., 2021), SimJoint (Shah et al., 2018), STAR (Mosig et al., 2020), SGD (Rastogi et al., 2020), OpenDialKG (Moon et al., 2019) and DuRecDial-2.0 (Liu et al., 2021).

Meanwhile, for knowledge-grounded dialogues, we drew upon dataset from SQA (Iyyer et al., 2017), SParC (Yu et al., 2019b), FeTaQA (Nan et al., 2022), MultiModalQA (Talmor et al., 2021), CompWebQ (Talmor and Berant, 2018), CoSQL (Yu et al., 2019a).

For open-domain dialogues, we sample dialogues from SODA (Kim et al., 2022a), ProsocialDialog (Kim et al., 2022b), Chitchat (Myers et al., 2020).

For each dialogue dataset, we sample at most 11k dialogues. Additionaly, we extracted around 11k dialogue turns from question-answering dialogues featured in RACE (Lai et al., 2017), NarrativeQA (Kočiskỳ et al., 2018), SQUAD (Rajpurkar et al., 2018), MCtest (Richardson et al., 2013), OpenBookQA (Mihaylov et al., 2018), MultiRC (Khashabi et al., 2018). Here, a dialogue turn refers to a pair consisting of a dialogue context and its corresponding system response. The rest datasets in DialogStudio are preserved for future evaluations and downstream fine-tuning.

For each dialogue during the training, we shape the available external knowledge into a string, which is included in dialogue context, and instruct the model to generate a dialogue response based on external knowledge. We use the format *Instruction \n <USER> user utterance <SYSTEM> system response <USER> ... <USER> user utterance \n <EXTERNAL KNOWLEDGE> supported knowledge* to train the model, where *<USER>*, *<SYSTEM>* and *<EXTERNAL KNOWLEDGE>* are special tokens.

We follow the public HuggingFace transformer code (Wolf et al., 2020; Wang et al., 2022) to train the model. For initializing our models, we adopt T5 (Raffel et al., 2020) and Flan-T5 (Longpre et al., 2023) as starting points to respectively establish DialogStudio-T5 and DialogStudio-Flan-T5. For the training of DialogStudio-Flan-T5, we exclude all translation-oriented tasks, limiting the sample size for each Flan task to a maximum of 150 examples. This leads to a cumulative total of 140,000 samples. We train the model up to 3 epochs with bfloat16 precision, a total batch size of 64. We set a constant learning rate 5e-5 and 3e-5 for the large model and the 3B model, respectively. Experiments are conducted using 16 A100 GPUs, each with 40GB of GPU memory.

4.2 Evaluation for Response Generation

Settings. We evaluate the performance on CoQA (Reddy et al., 2019) and MultiWOZ 2.2 (Zang et al., 2020). CoQA is a multi-turn conversational question answering dataset with 8k conversations about text passages from seven diverse domains. MultiWOZ 2.2 is one of the largest and most widely used multi-domain task-oriented dialogue corpora with more than 10000 dialogues. Each dialogue involves with one or more domains such as *Train, Restaurant, Hotel, Taxi,* and *Attraction*. The dataset is challenging and complex due to the multi-domain setting and diverse linguistic styles. Note that we exclude both datasets during the pre-training stage to prevent data leakage.

	CoQA		MultiWOZ	
	ROUGE-L	F1	ROUGE-L	F1
Flan-T5-3B (Longpre et al., 2023)	37.1	37.2	7.0	7.4
Flan-T5-Large (Longpre et al., 2023)	22.5	22.3	15.9	17.6
GODEL-Large (Peng et al., 2022)	43.2	43.3	18.5	19.3
DialogStudio-T5-Large	61.2	61.5	32.4	34.5
DialogStudio-Flan-T5-Large	63.3	63.5	33.7	35.9

Table 1: Zero-shot results on CoQA and MultiWOZ 2.2.

	CR (14 tasks)	DAR (7 tasks)	TE (27 tasks)	avg. (48 tasks)
OPT-30B (Zhang et al., 2022b)	21.3/1.1	35.2/4.1	40.3/0.9	32.3/2.0
OPT-IML-30B (Iyer et al., 2022)	37.4/41.6	51.4/51.8	54.7/47.8	47.9/47.1
OPT-175B (Zhang et al., 2022b)	21.0/4.2	37.1/16.8	41.6/2.2	33.3/7.7
OPT-IML-175B (Iyer et al., 2022)	39.0/49.8	**61.2**/60.2	54.3/51.0	**51.5**/53.6
Tk-INSTRUCT-11B (Wang et al., 2022)	32.3/**62.3**	51.1/**69.6**	**55.0/64.1**	46.1/**65.3**
Tk-INSTRUCT-3B (Wang et al., 2022)	38.4/51.3	45.7/58.5	48.4/52.8	44.2/54.2
DialogStudio-NIV2-T5-3B	**41.3**/59.8	57.5/63.7	52.3/55.1	50.4/59.5

Table 2: 0-shot/2-shot/5-shot ROUGE-L testing results on unseen datasets and unseen tasks. Results of baselines are reported by original papers. CR, DAR, and TE, avg. are abbreviations for Coreference Resolution, Dialogue Act Recognition, Textual Entailment, and average results, respectively.

For CoQA, we follow the original paper setting to answer question based on external passage. For MultiWOZ 2.2, we consider the lexicalized dialogue-act-to-response generation task where the model needs to consider both the dialogue context and the dialogue acts during generation. We follow the prompt from (Bang et al., 2023) to instruct models, i.e., *Continue the dialogue as a task-oriented dialogue system called SYSTEM. The answer of SYSTEM should follow the ACTION provided next while answering the USER's last utterance*.

We focus on zero-shot evaluation and report the ROUGE-L and F1 score (Miller et al., 2017), where ROUGE-L measures the longest common subsequence and F1 measures the Unigram F1 overlap between the prediction and ground-truth response.

Baselines. We consider GODEL (Peng et al., 2022) and Flan-T5 (Longpre et al., 2023) as our baselines. GODEL is a T5-based large pre-trained model for goal-oriented dialogues. It is pre-trained with 551M multi-turn Reddit dialogues and 5M knowledge-grounded and question-answering dialogues. Flan-T5 is an instruction-aware model. It is also initialized from T5 and pre-trained on the Flan collection, which consists of more than 1800 tasks and 400 datasets, including dialogue datasets.

Results. Table 1 depicts the results from both zero-shot and few-shot testing. Evidently, our models considerably surpass the baseline models in terms of zero-shot learning, exhibiting a robust generalized ability for response generation in a zero-shot scenario.

Flan-T5-3B, on the other hand, underperforms in the task of generating responses from dialog-acts. This model tends to produce incorrect dialog acts, unnatural utterances, or terminates with an empty end token. One explanation for this is that Flan-T5 models did not receive sufficient dialogue training during the instruction-training phase on the Flan collections. Comparisons between the performances of existing models before and after training on the unified dataset validate DialogStudio's usefulness.

4.3 Evaluation on Super-NaturalInstructions

Settings. NIV2 (Wang et al., 2022) introduces an instruction-tuning benchmark with more than 1600 tasks. We select 3 categories with 44 tasks from the held-out test set, which consists of 154

	MMLU		BBH
	0-SHOT	5-SHOT	3-SHOT
TK-INSTRUCT 11B (Wang et al., 2022)	-	41.1	32.9
LLAMA 13B (Touvron et al., 2023)	-	46.2	37.1
Vicuna 13B (Chiang et al., 2023)	-	49.7	37.1
Flan-T5-Large (Longpre et al., 2023)	41.5	41.9	37.1
Flan-T5-XL (Peng et al., 2022)	48.7	49.3	40.2
OPT-IML-Max 30B (Iyer et al., 2022)	46.3	43.2	31.3
DialogStudio-Flan-T5-Large	40.1	40.9	34.2
DialogStudio-Flan-T5-3B	48.3	47.8	40.3

Table 3: Test results on MMLU and BBH. Results come from original papers and InstructEval (Chia et al., 2023).

tasks, i.e., Coreference Resolution, Dialogue Act Recognition, and Textual Entailment. The selected tasks and datasets are unseen in the training stage. Specifically, we strictly follow all settings including metrics in (Wang et al., 2022), i.e., train models with instructions + 2 positive demonstrations and no negative demonstrations. We fine-tune DialogStudio-T5-3B on 756 training tasks.

Baselines. OPT (Zhang et al., 2022b) is a set of open decoder-only transformer models pre-trained on 180B tokens. OPT-IML (Iyer et al., 2022) is an instruction meta-learning model based on the OPT-IML bench with more than 1500 tasks. Tk-INSTRUCT (Wang et al., 2022) is initialized from T5 and further pre-trained based on NIV2 collections. Note that we neglect Flan-T5 because it trains with all the downstream datasets and tasks.

Results. Table 2 shows the 0-shot and 2-shot results on unseen datasets and unseen tasks. Based on the average performance on 48 tasks, DialogStudio-NIV2-T5-3B outperforms OPT-IML-175B by 5.9% on 2-shot learning with more than 50 times fewer model parameters, and it surpasses Tk-INSTRUCT-11B by 4.3% on 0-shot learning with more than 3 times fewer parameters. The performance demonstrates the strong generalization ability of DialogStudio model. Compared with Tk-INSTRUCT-3B, DialogStudio-NIV2-T5-3B achieves 6.2% and 5.3% improvements on 0-shot and 2-shot learning respectively. Given that both Tk-INSTRUCT and our DialogStudio-NIV2-T5-3B are fine-tuned from the T5 model, this improvement indicates the effectiveness of pre-training with our DialogStudio collection.

4.4 Evaluation on MMLU and BBH

Table 3 presents results on MMLU (Hendrycks et al., 2020) and Big Bench Hard (BBH) (Srivastava et al., 2022). When comparing the performance of DialogStudio-Flan-T5 with Flan-T5, we observe a minor decrease. However, this is accompanied by a significant improvement in dialogue relevant capabilities.

4.5 Evaluation on Alternative Benchmarks

DialogStudio encompasses not just public realistic dialogue datasets, but also those derived from or shared with ChatGPT, such as SODA (Kim et al., 2022a) and ShareGPT. Due to GPU constraints, we employ techniques like LoRA (Hu et al., 2021) to fine-tune llama (Touvron et al., 2023). When using equivalent datasets from DialogStudio, we observed performance comparable to other models, e.g., Vicuna (Chiang et al., 2023), on benchmarks like AlpacaEval (Dubois et al., 2023) and MT-Bench (Zheng et al., 2023). This demonstrates that DialogStudio caters to research interests in both specific datasets and generalized instruction tuning.

5 CONCLUSION

In this study, we have introduced DialogStudio, a comprehensive collection that aggregates more than 80 diverse dialogue datasets while preserving their original information. This aggregation not only represents a significant leap towards consolidating dialogues from varied sources but also offers a rich tapestry of conversational patterns, intents, and structures, capturing the nuances and richness of human interaction. Utilizing DialogStudio, we developed corresponding models, demonstrating superior performance in both zero-shot and few-shot learning scenarios. In the spirit of open research and advancing the field, we are committed to releasing DialogStudio to the broader research community.

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Appendix

Table 4 and Table 5 lists datasets included in DialogStudio. Initially, we present a partial list of these datasets. More and latest information are available in GitHub[2].

[2] https://github.com/salesforce/DialogStudio

NLU	NLU++ (Casanueva et al., 2022) BANKING77-OOS (Zhang et al., 2022a) BANKING77 (Casanueva et al., 2020) RESTAURANTS8K (Coope et al., 2020) CLINC150 (Larson et al., 2019) CLINC-Single-Domain-OOS-banking (Zhang et al., 2022a) CLINC-Single-Domain-OOS-credit_cards (Zhang et al., 2022a) HWU64 (Liu et al., 2019) SNIPS (Coucke et al., 2018) SNIPS-NER (Coucke et al., 2018) DSTC8-SGD (Coope et al., 2020) TOP (Gupta et al., 2018) TOP-NER (Gupta et al., 2018) ATIS-NER (Hemphill et al., 1990) ATIS (Hemphill et al., 1990) MIT-MOVIE (Liu et al., 2013) MIT-RESTAURANT (Liu et al., 2013)
TOD	KVRET (Eric et al., 2017) AirDialogue (Wei et al., 2018) DSTC2-Clean (Mrkšić et al., 2017) CaSiNo (Chawla et al., 2021) FRAMES (El Asri et al.) WOZ2.0 (Mrkšić et al., 2017) CraigslistBargains (He et al., 2018) Taskmaster1 (Byrne et al., 2019) Taskmaster2 (Byrne et al., 2019) Taskmaster3 (Byrne et al., 2019) ABCD (Chen et al., 2021a) MulDoGO (Peskov et al., 2019) BiTOD (Lin et al., 2021) SimJointGEN (Shah et al., 2018) SimJointMovie (Shah et al., 2018) SimJointRestaurant (Shah et al., 2018) STAR (Mosig et al., 2020) SGD (Rastogi et al., 2020) MultiWOZ2_1 (Eric et al., 2020) MultiWOZ2_2 (Zang et al., 2020) MultiWOZ2_2+ (Qian et al., 2021) HDSA-Dialog (Chen et al., 2021a) MS-DC (Li et al., 2018b) GECOR (Quan et al., 2019) Disambiguation (Qian et al., 2022) MetaLWOZ (Lee et al., 2019) KETOD (Chen et al., 2022b) MuDoCo (Martin et al., 2020)

Table 4: List of datasets included in DialogStudio (a).

KG-Dial	SQA (Iyyer et al., 2017) SParC (Yu et al., 2019b) FeTaQA (Nan et al., 2022) MultiModalQA (Talmor et al., 2021) CompWebQ (Talmor and Berant, 2018) CoSQL (Yu et al., 2019a) CoQA (Reddy et al., 2019) Spider (Yu et al., 2018) ToTTo (Parikh et al., 2020) WebQSP (Yih et al., 2016) WikiSQL (Zhong et al., 2017) WikiTQ (Pasupat and Liang, 2015) DART (Nan et al., 2021) GrailQA (Gu et al., 2021) HybridQA (Chen et al., 2020) MTOP (Chen et al., 2020) UltralChat-Assistance (Ding et al., 2023) Wizard_of_Wikipedia (Dinan et al., 2018) Wizard_of_Internet (Komeili et al., 2022)
Dial-Sum	TweetSumm (Feigenblat et al., 2021) SAMSum (Gliwa et al., 2019) DialogSum (Chen et al., 2021b) AMI (Kraaij et al., 2005; Rennard et al., 2023) ICSI (Janin et al., 2003) QMSum (Zhong et al., 2021) MediaSum (Zhu et al., 2021) ECTSum (Mukherjee et al., 2022) SummScreen_ForeverDreaming (Chen et al., 2022a) SummScreen_TVMegaSite (Chen et al., 2022a) CRD3 (Rameshkumar and Bailey, 2020) ConvoSumm (Fabbri et al., 2021)
Open-Domain	ChitCHAT (Myers et al., 2020) SODA (Kim et al., 2022a) Prosocial (Kim et al., 2022b) HH-RLHF (Bai et al., 2022) Empathetic (Rashkin et al., 2019) ConvAI2 (Dinan et al., 2019) AntiScam (Li et al., 2020) ShareGPT (Zheng et al., 2023) PLACES3.5 (Chen et al., 2023)
Conv-Rec	SalesBot (Chiu et al., 2022) Redial (Li et al., 2018a) Inspired (Hayati et al., 2020) DuRecDial 2.0 (Liu et al., 2021) OpendialKG (Moon et al., 2019)

Table 5: List of datasets included in DialogStudio (b).